26.06.25

OPENCAMPUS.sh

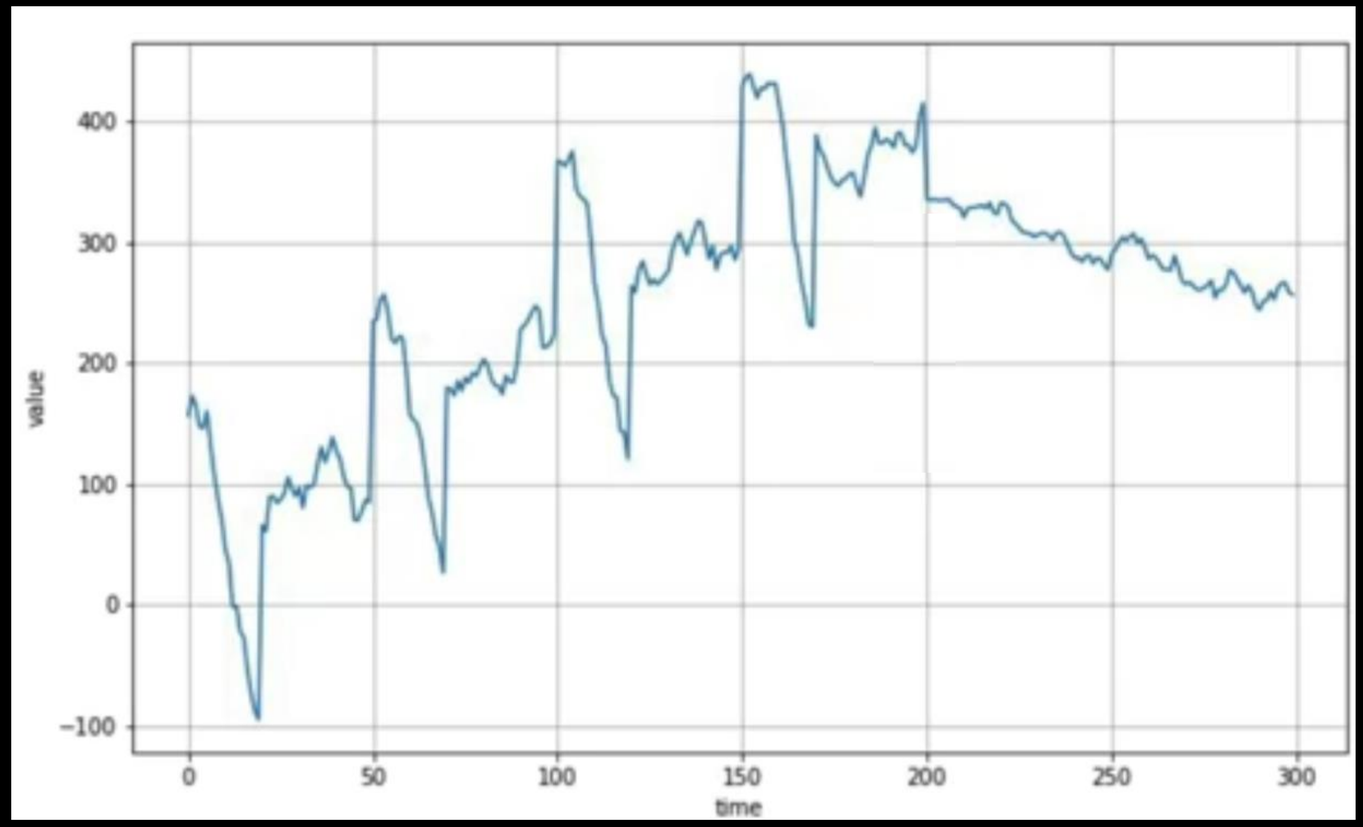
Introduction to Data Science and Machine Learning

TIME SERIES ANALYSES

- **Patterns in Time Series**
- **Non-Stationarity**
- **Baseline Models und Naïve Forecasting**
- **Project Presentation**
- ***And now?***

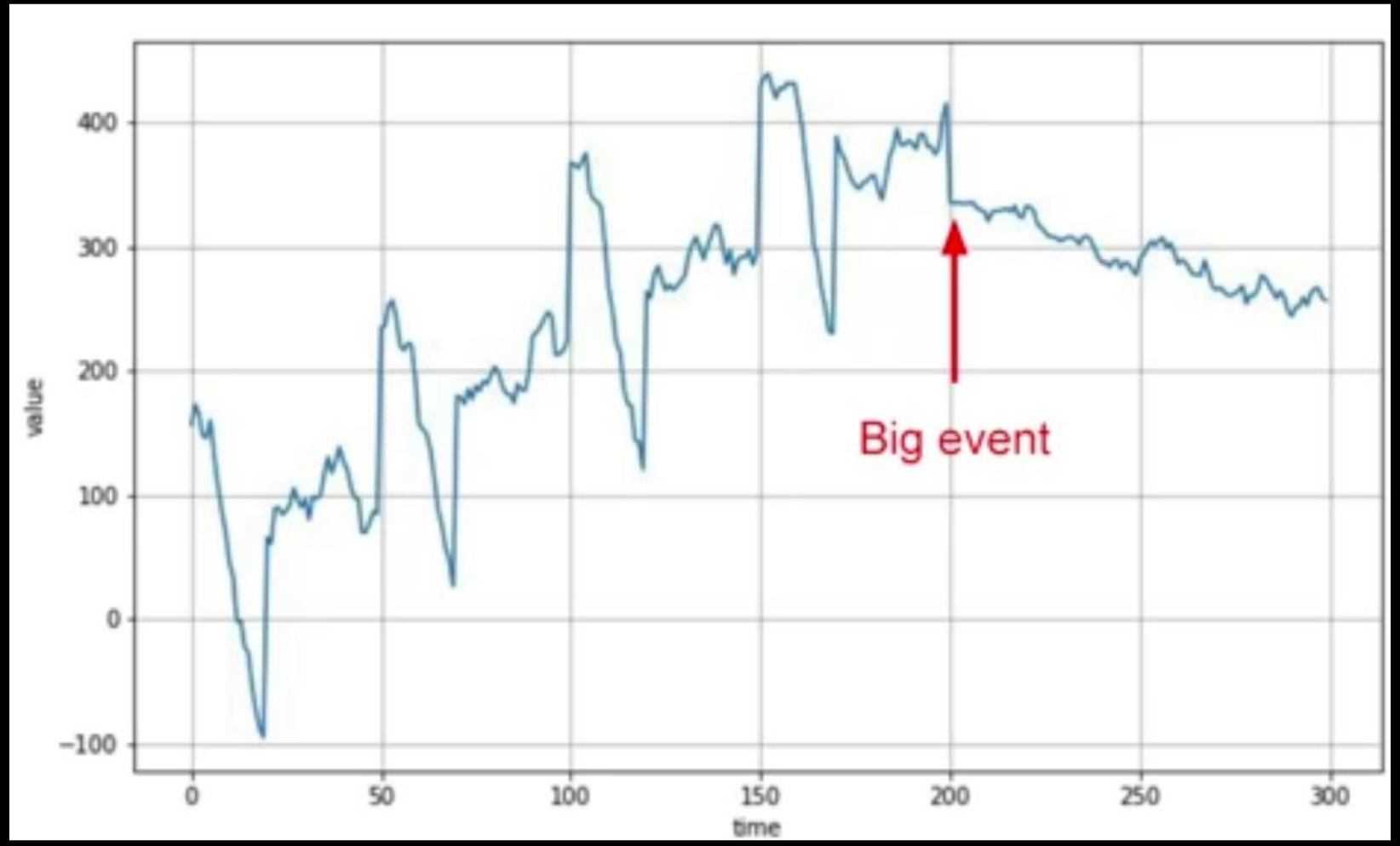
DISCUSSION

What different types of patterns can be identified in the displayed figure?



DeepLearning.AI. (o. J.). Common patterns in time series—Sequences and Prediction. Abgerufen 7. Januar 2021, von Coursera website: https://www.coursera.org/lecture/tensorflow-sequences-time-series-and-prediction/common-patterns-in-time-series-dWnOy

NON-STATIONARITY

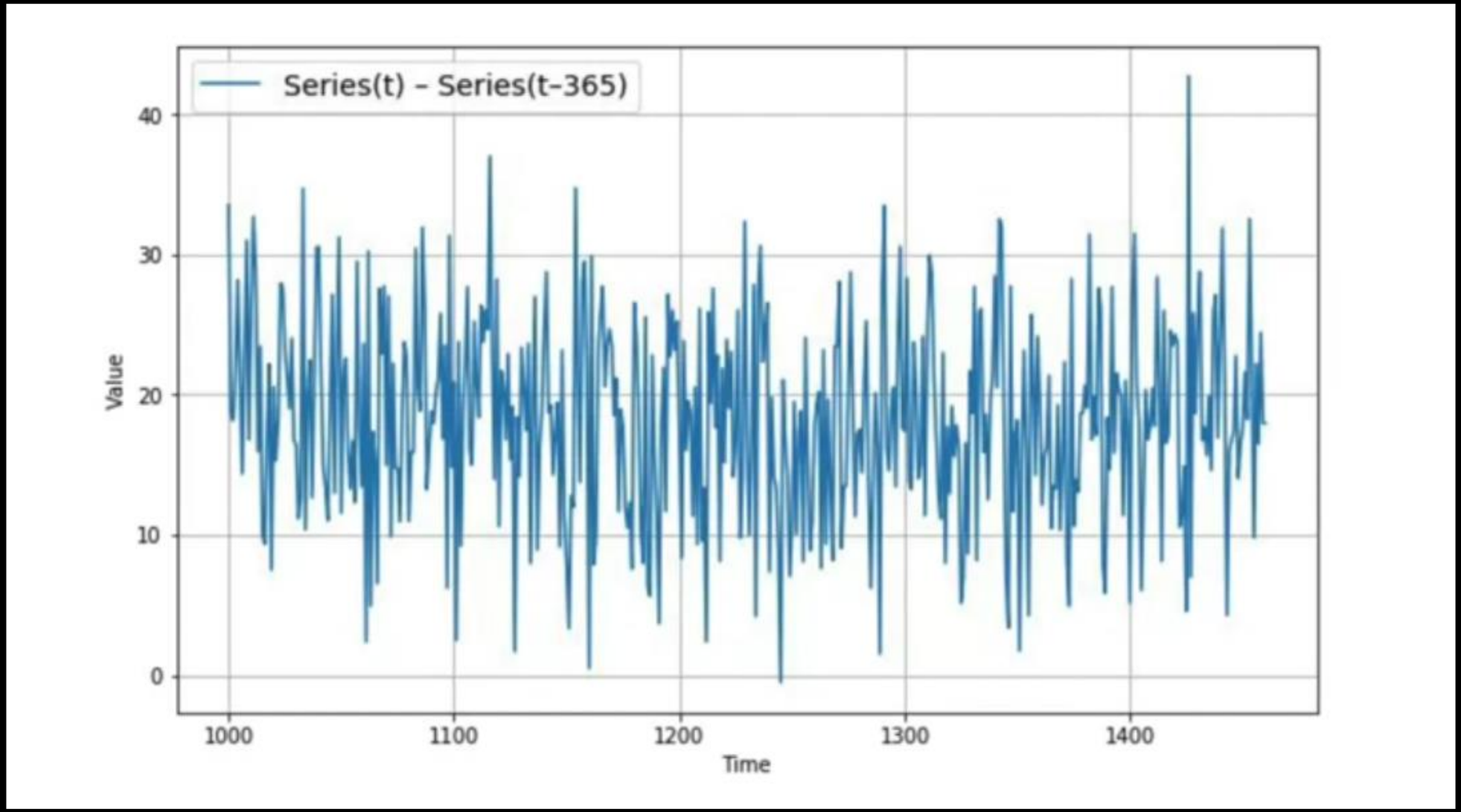


DeepLearning.AI. (o. J.). Common patterns in time series—Sequences and Prediction. Abgerufen 7. Januar 2021, von Coursera website: https://www.coursera.org/lecture/tensorflow-sequences-time-series-and-prediction/common-patterns-in-time-series-dWnOy

PATTERNS IN TIME SERIES

- Trends
- Seasonality
- Noise
- Autocorrelation (serial correlation)

DIFFERENCING



DIFFERENCING

- Subtracting the previous observation from the current observation
- A method for removing trends, seasonality, or other potential autocorrelations in time series
- Helps to reduce autocorrelation and obtain stationary time series

„LAG“

- The difference between consecutive observations is called the Lag-1 difference.
- More generally, a Lag-k difference refers to the difference between an observation and the observation k time steps earlier.
- The lag value should be chosen based on the temporal structure of the data.
- For time series with a seasonal component, the lag is typically set to match the length of the seasonal cycle (e.g., lag = 12 for monthly data with yearly seasonality).

BASELINE MODELS

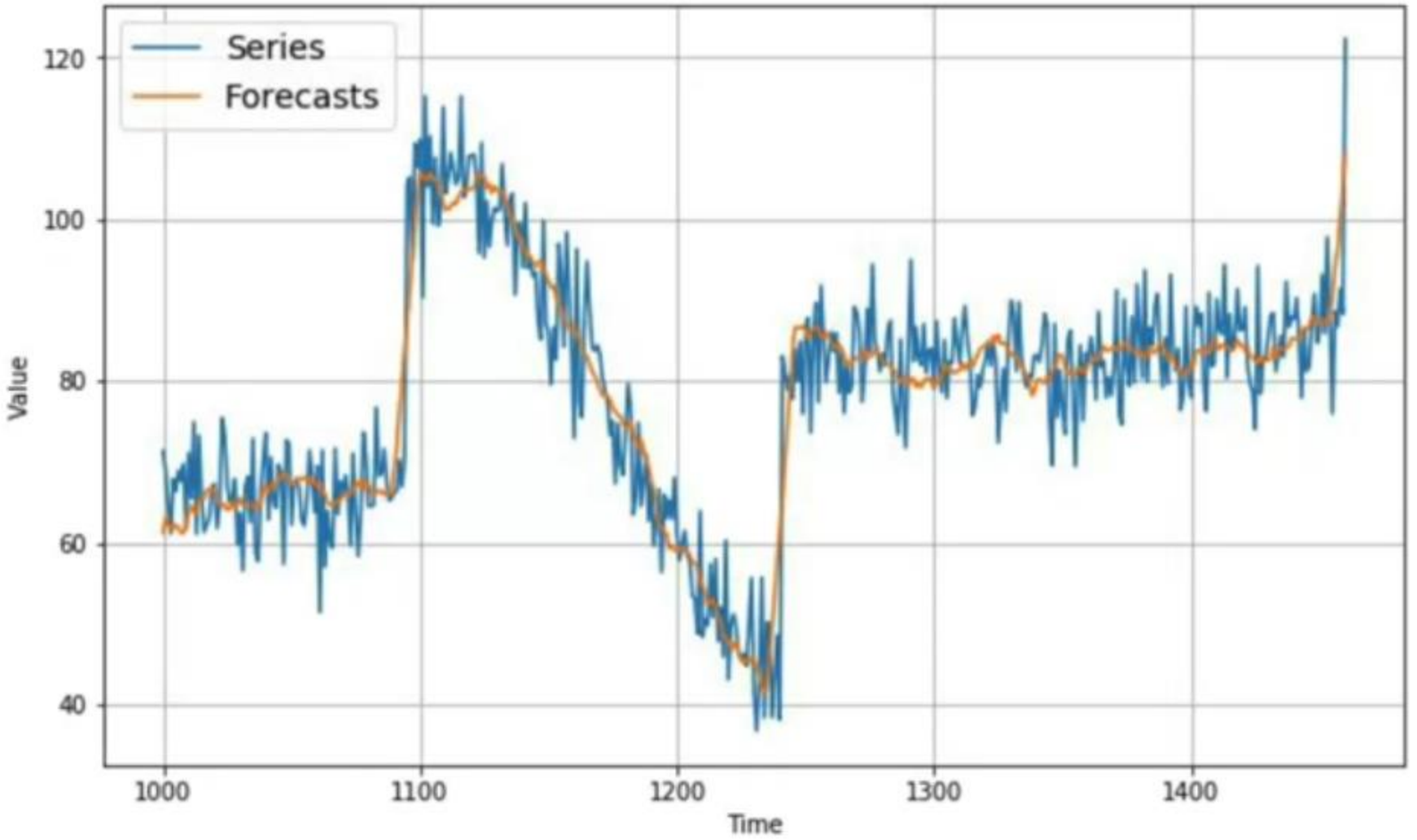
- Metrics like MAPE or RMSE alone often don't reveal how well a model performs.
- Always compare results to a baseline for meaningful evaluation.

Typical Baselines:

- Previous models on the same dataset
- Models on similar datasets
- For time series: Naïve Forecasting

NAÏVE FORECASTING

- Simple baseline model for time series analysis
- Forecast = last observed value
- Seasonal Naïve Forecasting:
  Forecast = last observed value from the same seasonal period



Forecasts = trailing moving average of differenced series + centered moving average of past series (t – 365)

DISCUSSION

In practice, the focus is often on forecasting the next day, not the whole year.

- **Which time series features help most with short-term forecasts?**

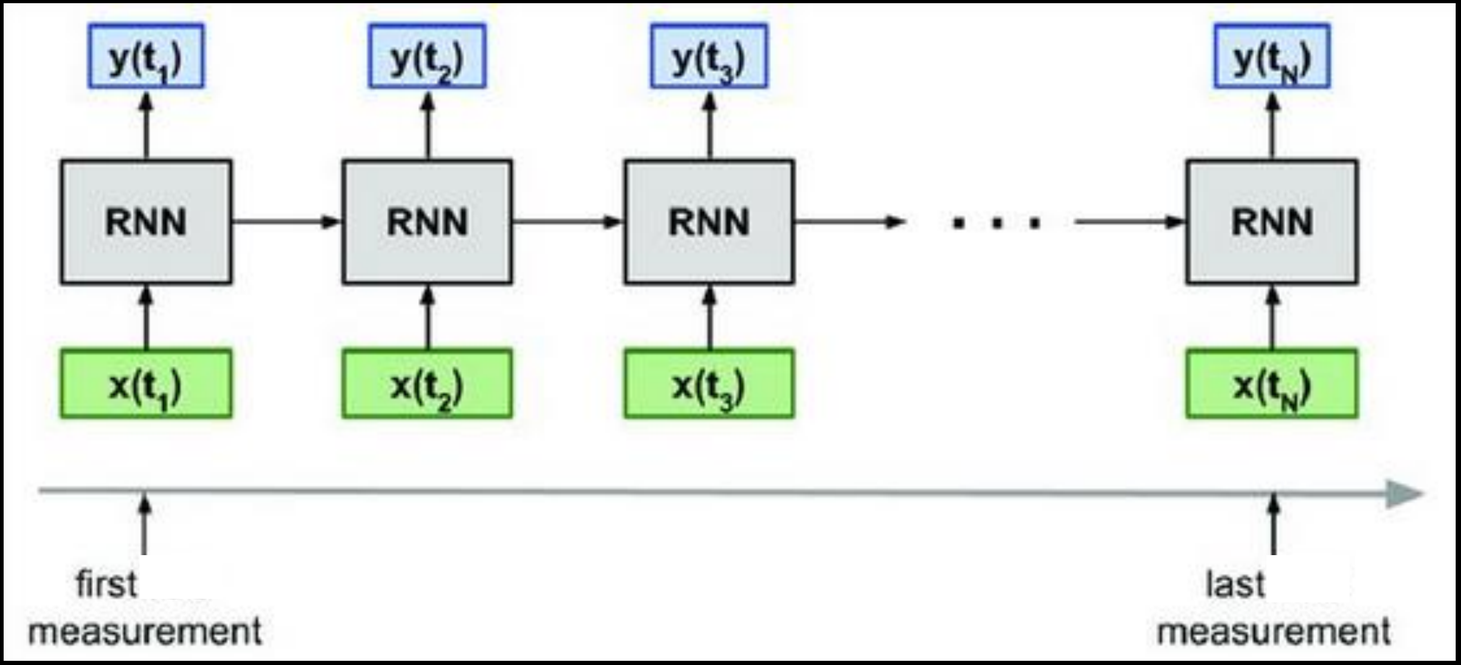
BREAKOUT

- Discuss options to incorporate time series information in your dataset

- Try implementing it

RECURRENT NEURAL NET (RNN)

- In an RNN layer, the nodes within a layer are interconnected (recurrently connected).



PROJECT PRESENTATION

Presentation (Powerpoint, Keybote or similar)

Prepare an 8 to 10-minute presentation including:

- Your team members' names on the title slide
- List and brief description of self-created variables
- Bar charts with confidence intervals for two self-created variables
- Linear model optimization: model equation and adjusted R^2
- Type of missing value imputation used
- Neural network optimization:
  - Source code defining the neural network
  - Loss function plots for training and validation sets
  - MAPE scores for the overall validation set and each product group
- Highlight "Worst Fail" and "Best Improvement" cases

Each team member should have a part in the presentation!

Important Notes

- **Presentation duration: approx. 8 minutes per team**
- **Every team member should have a part**
- **Use PowerPoint, Keynote, or similar**

Deadlines

- **By 4 PM next week:**
  Generate predictions for the Kaggle competition test dataset using your best model and upload them
- **By July 31, 2025:**
  - **Add your presentation to the repository**
  - **Set the repository to public**
  - **Complete the repository as described in the READMEs**
  - **One team member uploads the main README to the EduHub platform as instructed**

TASKS

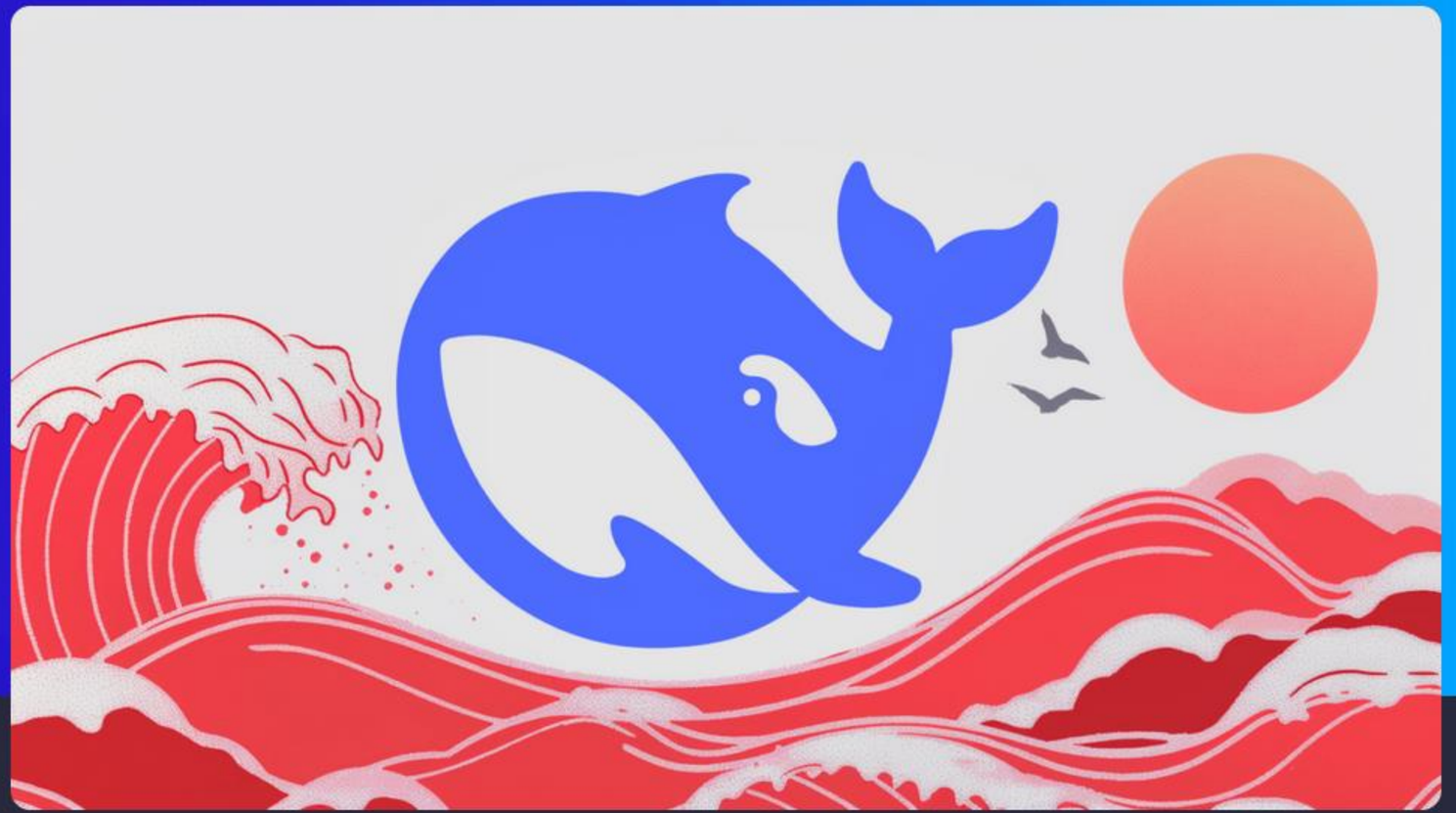
- Using your best model, generate predictions for the Kaggle competition test dataset and upload them there.

- Prepare your final presentation (see the guidelines for Week 10 here).

HOW CAN I MOVE ON IN MY MACHINE LEARNING JOURNEY

KI-Forschung 27. Dezember 2024

Deepseek-V3: Ein 5,6-Millionen-Dollar-Wunder aus China mischt die KI-Elite auf



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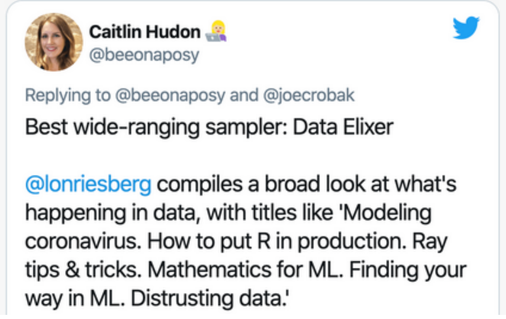
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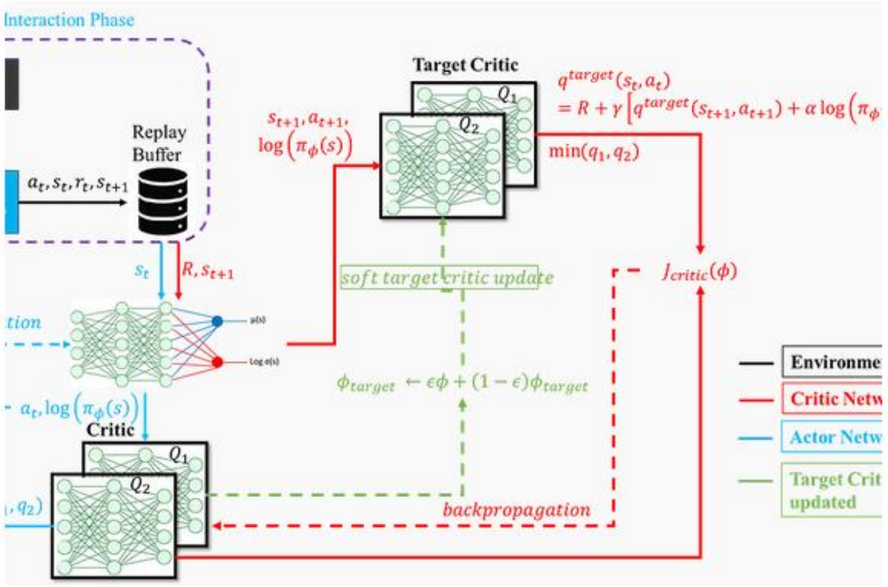
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Heiß

Community-Highlights

[D] Simple Questions Thread
0 Votes • 4 Kommentare

[D] Monthly Who's Hiring and Who wants to be Hired?
31 Votes • 2 Kommentare

u/alexsht1 • vor 2 Std.

[R] Numerical features with factorization machines

Happy to share our recent TMLR paper, "Function Basis Encoding of Numerical Features in Factorization Machines", by Alex Shtoff, Elie Abboud, Rotem Stram, and Oren Somekh. This paper proposes an interesting insight into the interplay between Factorization Machines (FMs), and feature encoding using basis functions, in the context of recommender systems. The same interplay with linear models is an old classic, and most of us have learned in our ML 101 courses. Polynomial regression is one of them - we encode a feature x using the standard polynomial basis $\{1, x, x^2, ...\}$. FMs are family of models that model a quadratic polynomial...

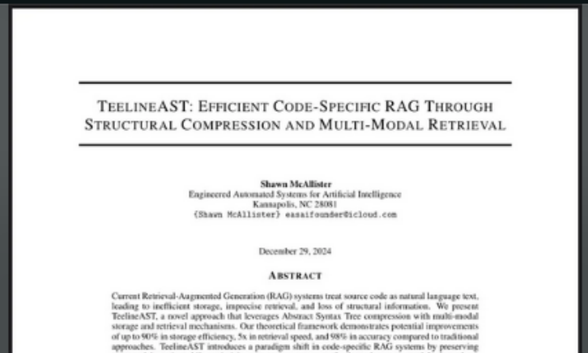
Research

9 | 0 | Teilen

u/stonedoubt • vor 12 Std.

[R]AST+Shorthand+HybridRag

Research



TeelineAST: Efficient Code-Specific RAG Through Structural Compression and Multi-Modal Retrieval

Shawn McAllister
Engineered Automated Systems for Artificial Intelligence
Kannapolis, NC 28081
{Shawn McAllister} eaasifounder@icloud.com

December 29, 2024

Abstract

Current Retrieval-Augmented Generation (RAG) systems treat source code as natural language text, leading to inefficient storage, imprecise retrieval, and loss of structural information. We present TeelineAST, a novel approach that leverages Abstract Syntax Tree compression with multi-modal storage and retrieval mechanisms. Our theoretical framework demonstrates potential improvements of up to 90% in storage efficiency, 5x in retrieval speed, and 98% in accuracy compared to traditional approaches. TeelineAST introduces a paradigm shift in code-specific RAG systems by preserving

Machine Learning
Beginners -> /r/mlquestions , AGI -> /r/singularity, career advices -> /r/cscareerquestions, datasets -> r/datasets

Erstellt am 29. Juli 2009
Öffentlich

2,9 Mio. Mitglieder | **128** Online | **Oberste 1 %** Nach Größe sortieren

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Langzeitmitglied
1 freigeschaltet | Alle anzeigen

REGELN

1. Be nice: no offensive behavior, insults or attacks
2. Make your post clear and comprehensive
3. Posts without appropriate tag in title will be removed
4. Beginner or career related questions go elsewhere
5. Non-arxiv link posts only allowed on weekends (must be demos)*
6. Beginner's tutorials and projects go elsewhere
7. Quality Contribution
8. Limit self-promotion

▲ Safe Superintelligence Inc. (ssi.inc)

1023 points by nick_pou 16 hours ago | hide | past | favorite | 860 comments

add comment

▲ insane_dreamer 15 hours ago | next [–]

I understand the concern that a "superintelligence" will emerge that will escape its bounds and threaten humanity. That is a risk.

My bigger, and more pressing worry, is that a "superintelligence" will emerge that does not escape its bounds, and the question will be which humans control it. Look no further than history to see what happens when humans acquire great power. The "cold war" nuclear arms race, which brought the world to the brink of (at least partial) annihilation, is a good recent example.

Quis custodiet ipsos custodes? -- That is my biggest concern.

Update: I'm not as worried about Ilya et al as commercial companies (including formerly "open" OpenAI) discovering AGI.

reply

▲ benreesman 5 hours ago | parent | next [–]

It's just clearly military R&D at this point.

And it's not even a little bit controversial that cutting edge military R&D is classified in general and to an extreme in wartime.

The new thing is the lie that it's a consumer offering. What's new is giving the helm to shady failed social network founders with no accountability.

These people aren't retired generals with combat experience. They aren't tenured professors at Princeton IAS on a Nobel shortlist and encumbered by TS clearance.

They're godawful almost ran psychos who never built anything that wasn't extractive and owe their position in the world to pg's partisanship 15 fucking years ago.

reply

▲ yayr 0 minutes ago | root | parent | next [–]

most technology is dual or multiple use, starting with a rock or knife...

so it is up to the fabric of our society and everyone involved in dealing with the technology, how the rules and boundaries are set.

that there will be military use is obvious. However, it is naive to think one can avoid military use by others by not enabling oneself for it.

reply

▲ sgregnt 3 hours ago | root | parent | prev | next [–]

To me it is not clear at all, can you please elaborate why you make such a strong claim?

LINKEDIN

- Philipp Schmid
- Lior Sinclair
- Hugging Face
- NVIDIA
- …


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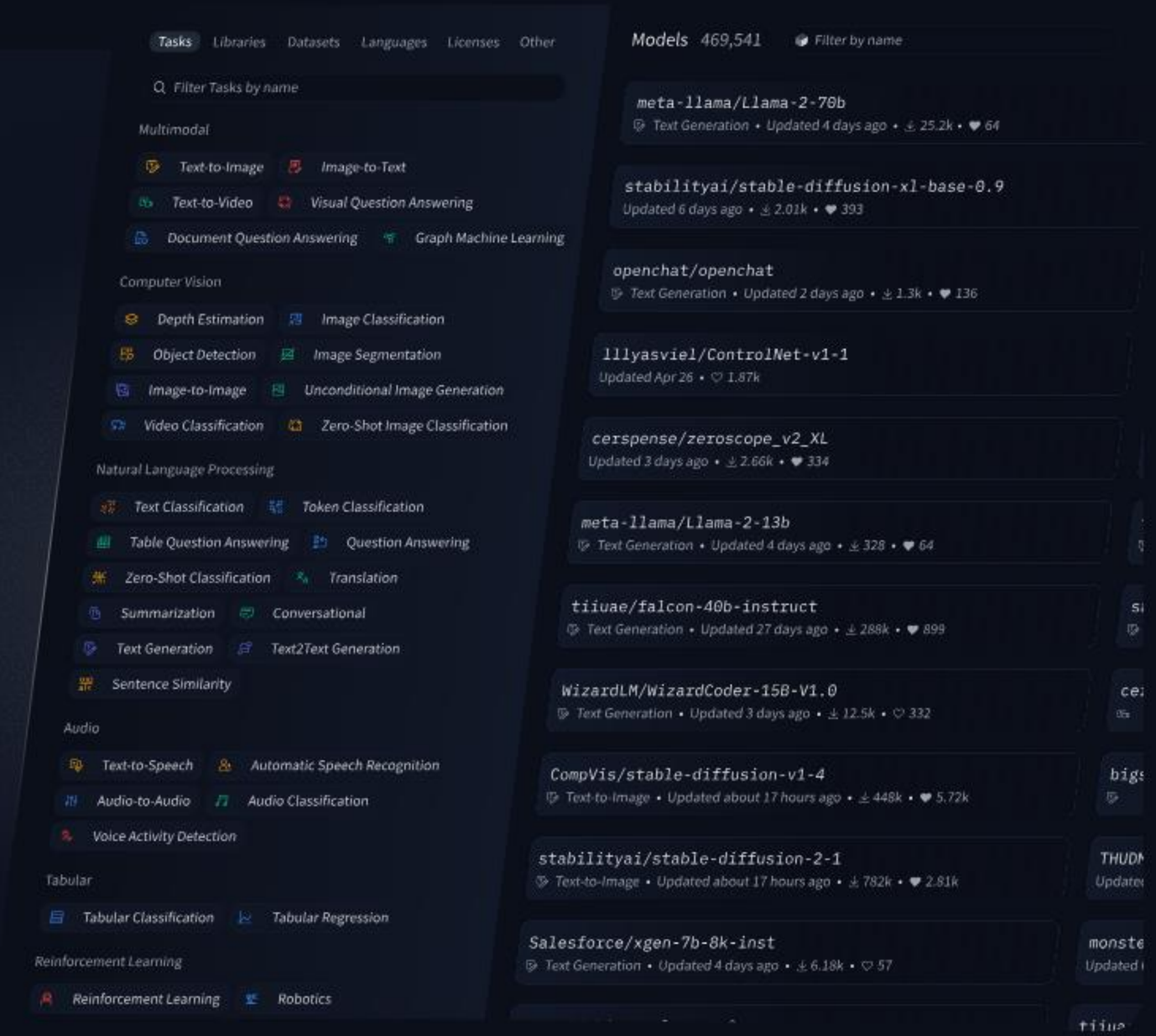
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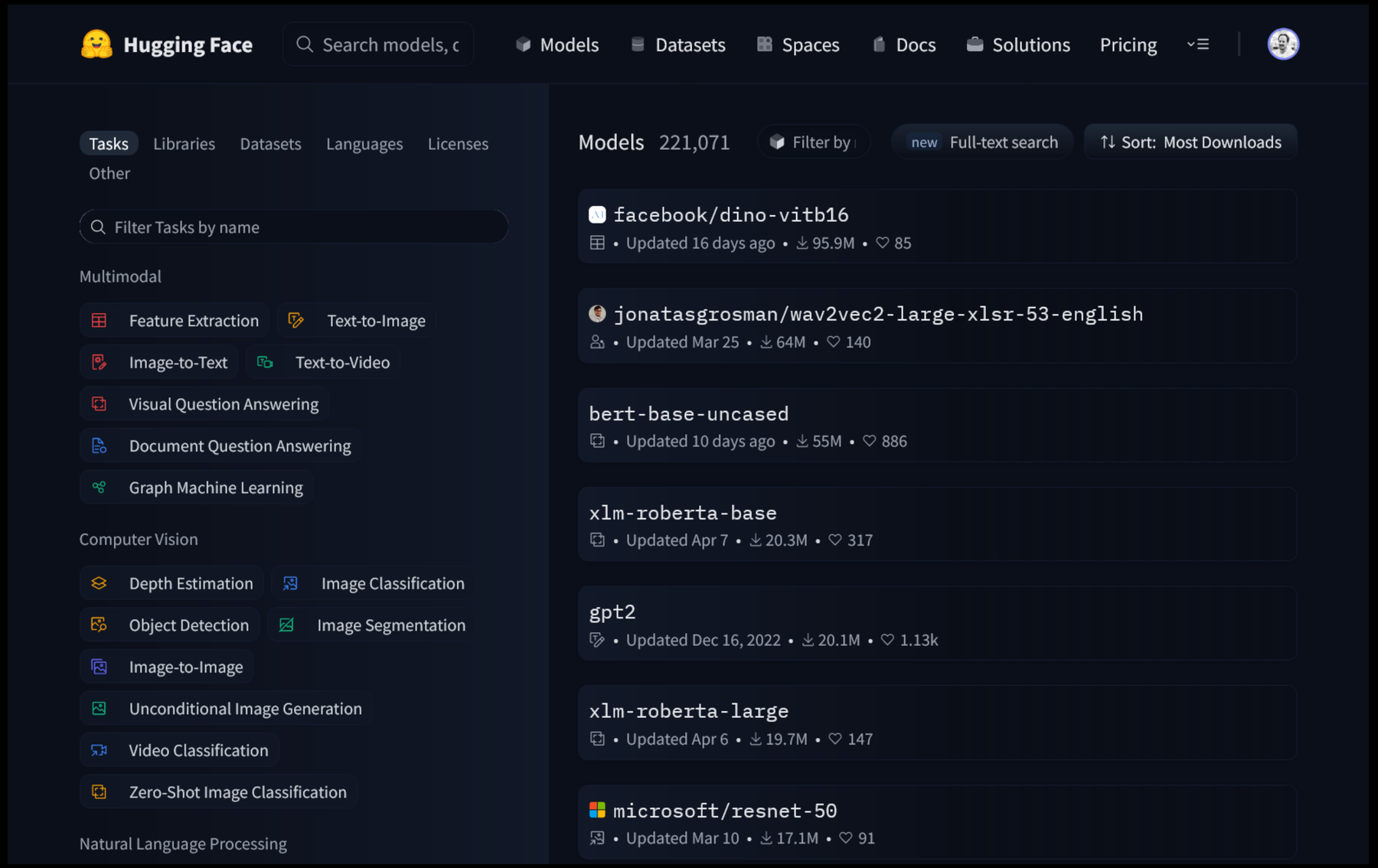
The AI community building the future.

The platform where the machine learning community collaborates on models, datasets, and applications.

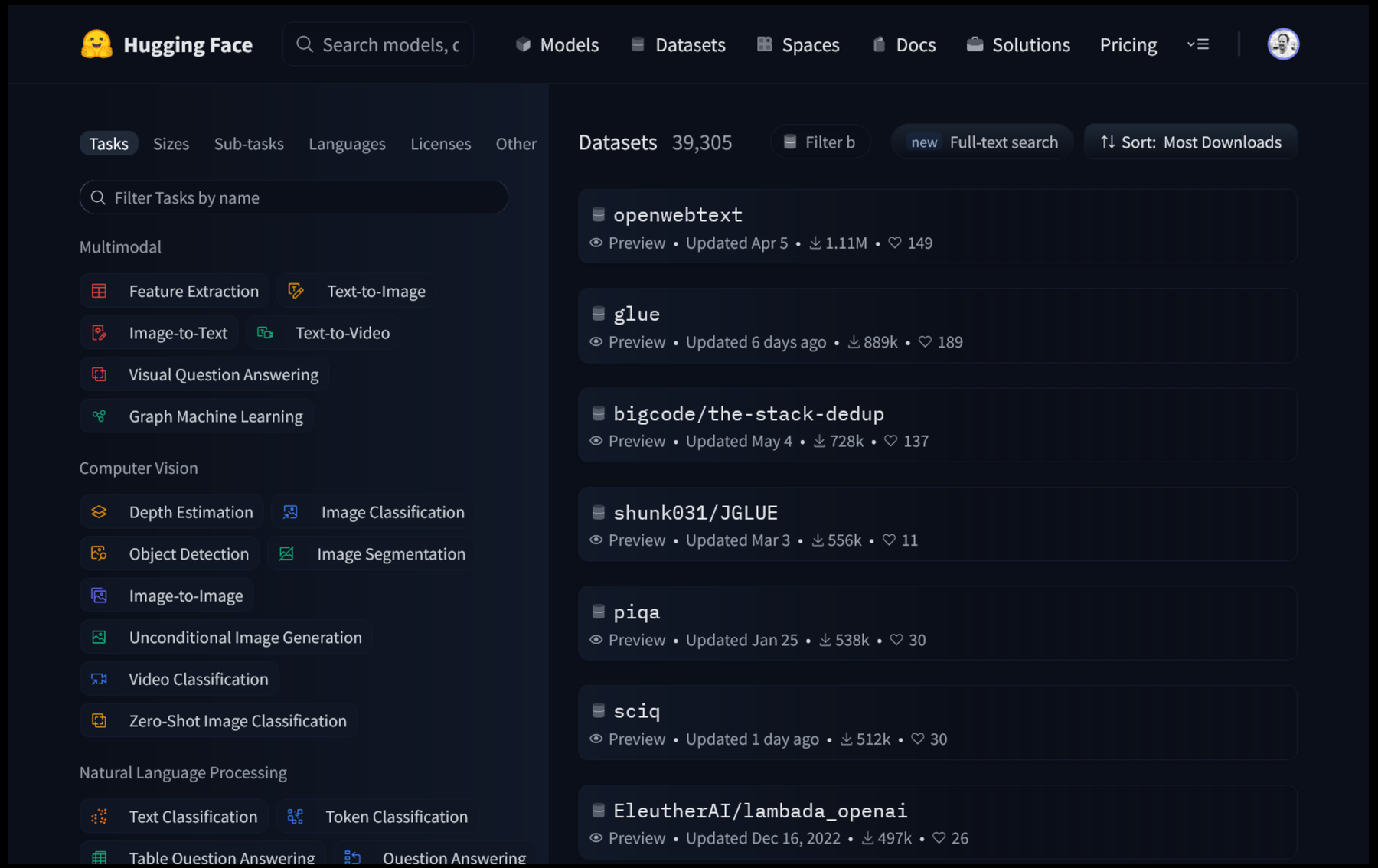
Explore AI Apps or Browse 1M+ models



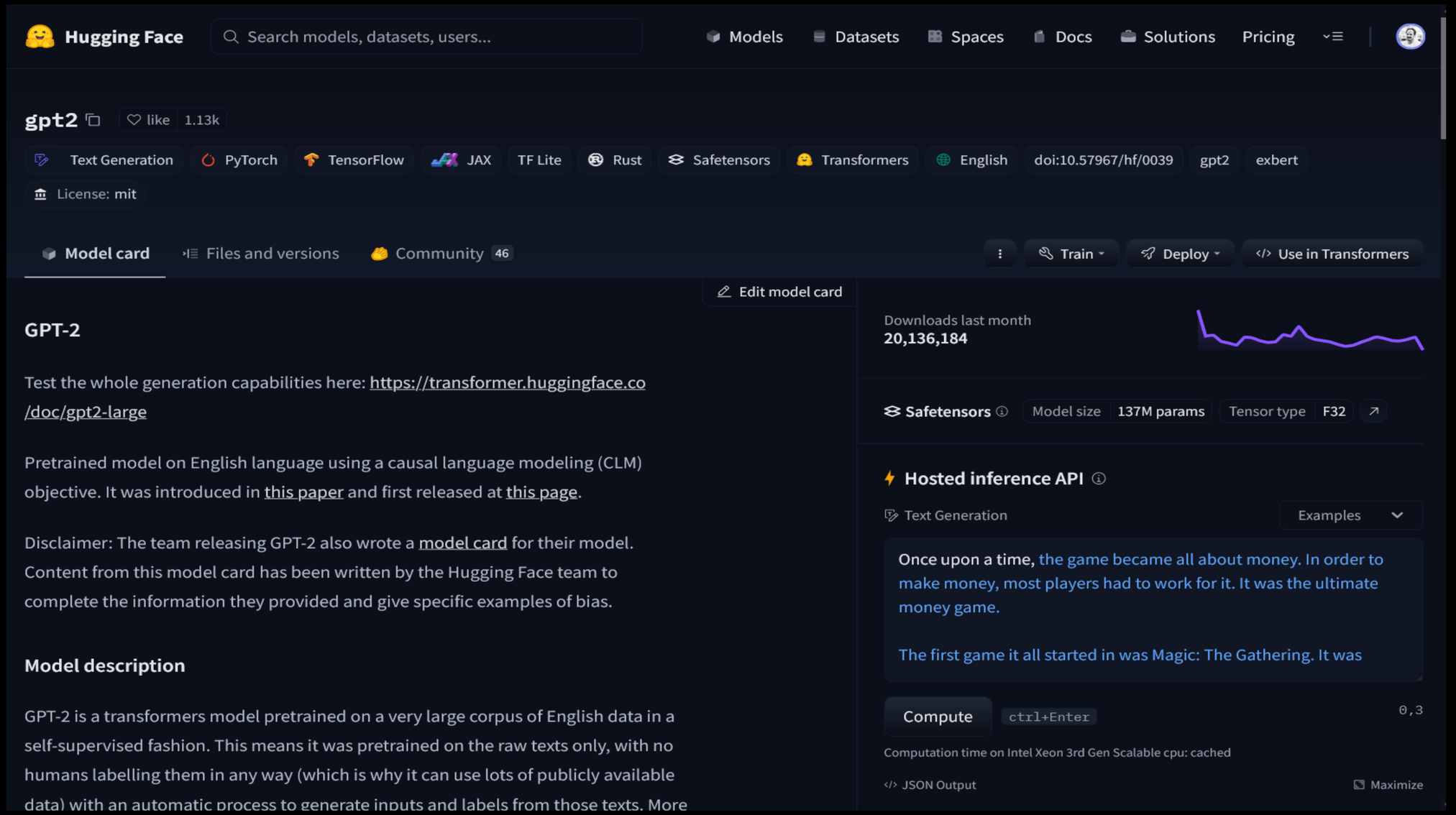
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Feature Extraction
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Image-to-Text
Text-to-Video
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Document Question Answering
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MODEL CARD


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Model card
Files and versions
Community 46
Train
Deploy
Use in Transformers
Edit model card
GPT-2
Test the whole generation capabilities here: https://transformer.huggingface.co/doc/gpt2-large
Pretrained model on English language using a causal language modeling (CLM) objective. It was introduced in this paper and first released at this page.
Disclaimer: The team releasing GPT-2 also wrote a model card for their model. Content from this model card has been written by the Hugging Face team to complete the information they provided and give specific examples of bias.
Model description
GPT-2 is a transformers model pretrained on a very large corpus of English data in a self-supervised fashion. This means it was pretrained on the raw texts only, with no humans labelling them in any way (which is why it can use lots of publicly available data) with an automatic process to generate inputs and labels from those texts. More
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20,136,184
Safetensors
Model size 137M params
Tensor type F32
Hosted inference API
Text Generation
Examples
Once upon a time, the game became all about money. In order to make money, most players had to work for it. It was the ultimate money game.
The first game it all started in was Magic: The Gathering. It was
Compute
ctrl+Enter
0,3
Computation time on Intel Xeon 3rd Gen Scalable cpu: cached
JSON Output
Maximize

PIPELINE FUNCTION

```
from transformers import pipeline

classifier = pipeline("sentiment-analysis")
classifier("I've been waiting for a HuggingFace course my whole life.")
```

```
[{'label': 'POSITIVE', 'score': 0.9598047137260437}]
```

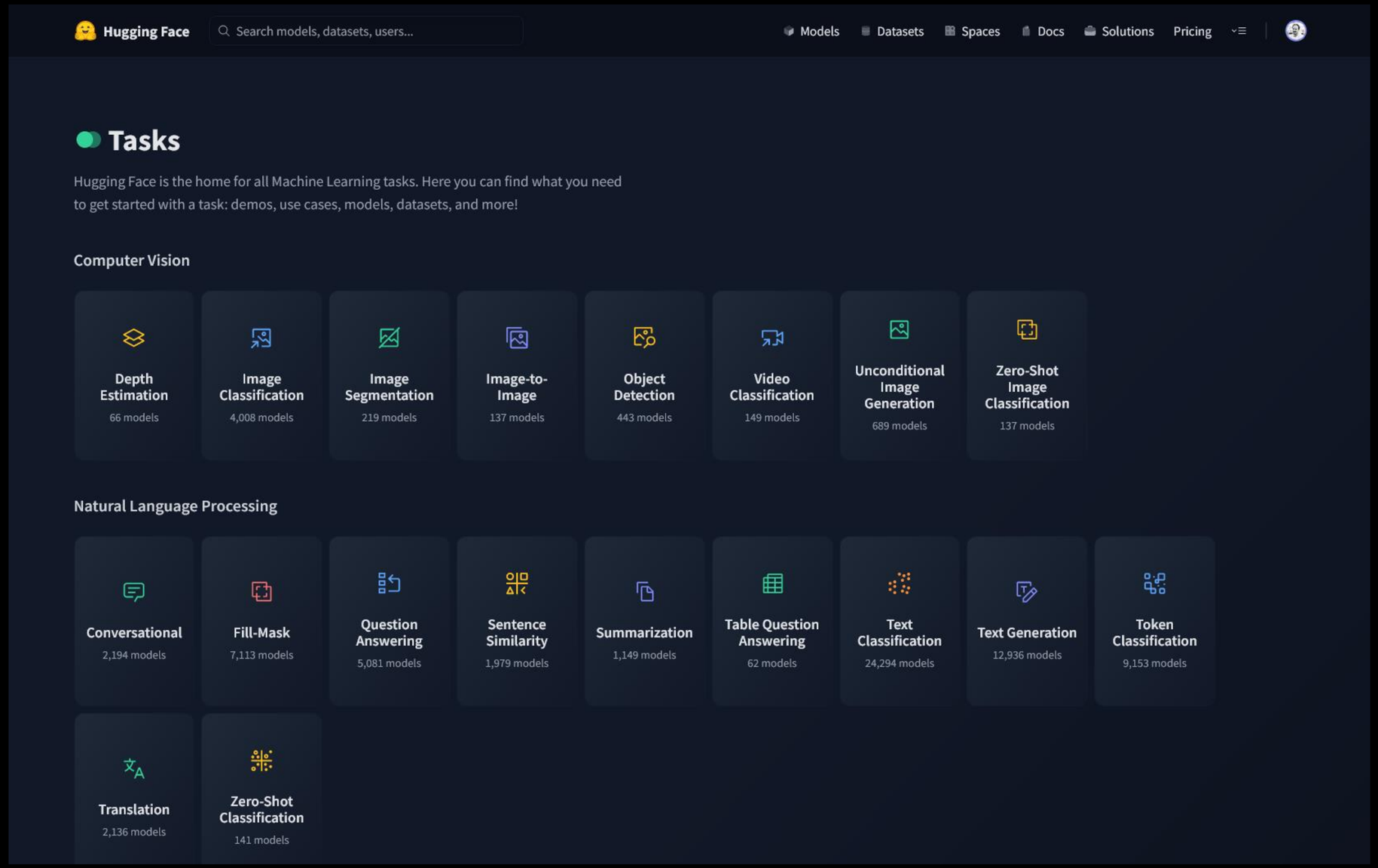
TASK IDENTIFIER

task (`str`) –

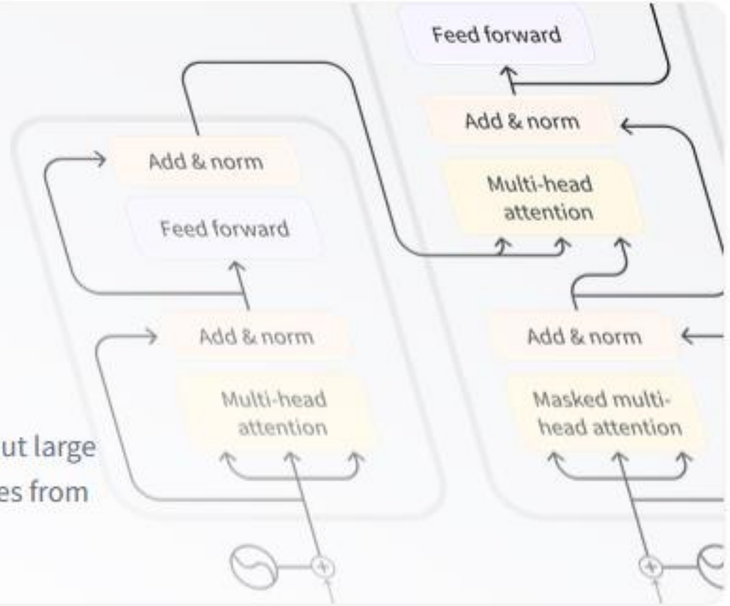
The task defining which pipeline will be returned. Currently accepted tasks are:

- "feature-extraction": will return a `FeatureExtractionPipeline`
- "sentiment-analysis": will return a `TextClassificationPipeline`
- "ner": will return a `TokenClassificationPipeline`
- "question-answering": will return a `QuestionAnsweringPipeline`
- "fill-mask": will return a `FillMaskPipeline`
- "summarization": will return a `SummarizationPipeline`
- "translation_xx_to_yy": will return a `TranslationPipeline`
- "text-generation": will return a `TextGenerationPipeline`

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66 models
Image Classification
4,008 models
Image Segmentation
219 models
Image-to-Image
137 models
Object Detection
443 models
Video Classification
149 models
Unconditional Image Generation
689 models
Zero-Shot Image Classification
137 models
Natural Language Processing
Conversational
2,194 models
Fill-Mask
7,113 models
Question Answering
5,081 models
Sentence Similarity
1,979 models
Summarization
1,149 models
Table Question Answering
62 models
Text Classification
24,294 models
Text Generation
12,936 models
Token Classification
9,153 models
Translation
2,136 models
Zero-Shot Classification
141 models

Learn



LLM Course

This course will teach you about large language models using libraries from the HF ecosystem

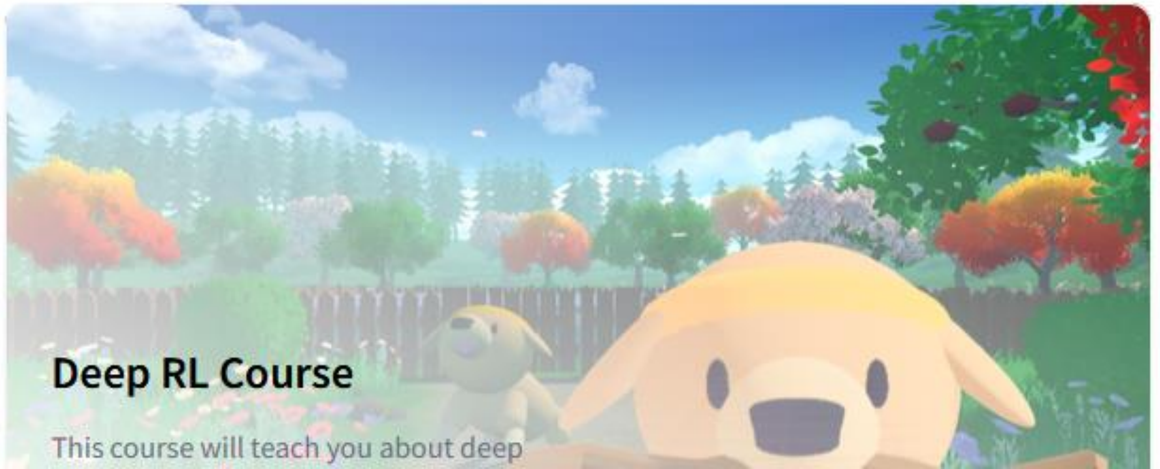


MCP Course

This course will teach you about Model Context Protocol

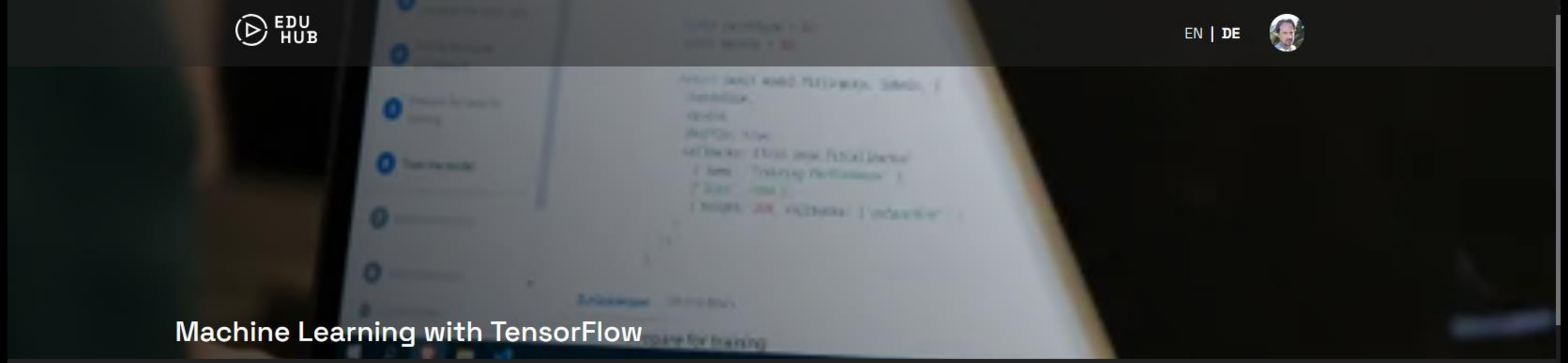


Agents Course



Deep RL Course

This course will teach you about deep



Machine Learning with TensorFlow

DONNERSTAG 18:15 - 20:00

Get hands-on experience in applying machine learning techniques with TensorFlow.

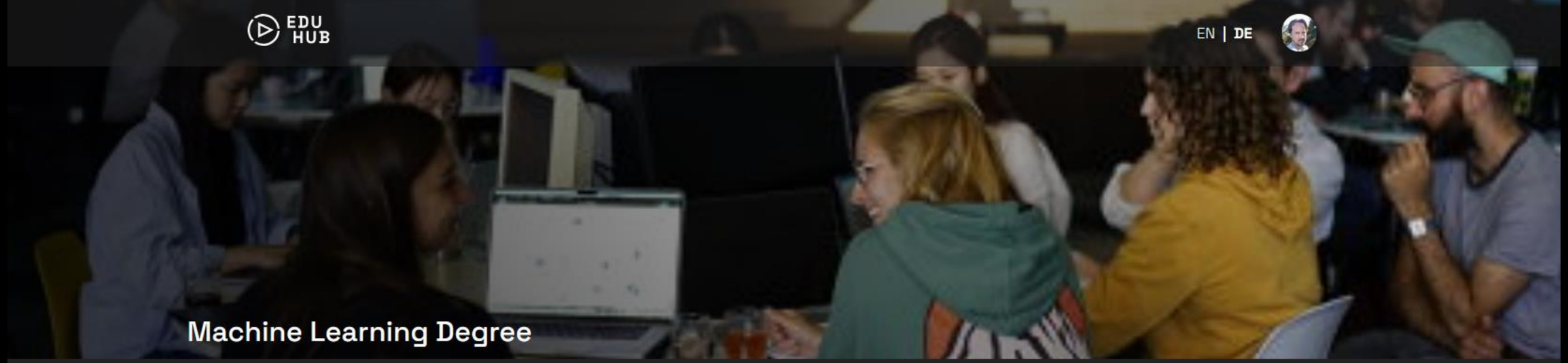
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Du wirst lernen

- Mastering best practices for TensorFlow, a popular open-source machine learning framework, to train neural networks
- Handling real-world image data and exploring strategies to avoid overfitting, including techniques like augmentation and dropout
- Building a system capable of processing natural language
- Applying Recurrent Neural Networks (RNNs), Gated Recurrent Units (GRUs), and Long Short-Term Memory (LSTMs) networks to train models using text and time-series data

DONNERSTAG	ECTS
18:15 - 20:00	5
HEIDE + KIEL + ONLINE	ENGLISCH

Kurstermine



Machine Learning Degree

After completing this program you will have a solid understanding of machine learning and will be able to implement your own state of the art machine learning projects.

Die Bewerbungsfrist ist leider abgelaufen. Bleibe immer auf dem Laufenden und verpasse nicht unser nächstes Kursprogramm, Events, Jobs und was es sonst noch Neues gibt: ZUM NEWSLETTER

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Kiel, Deutschland

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